

Vatsal Jha

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Bangalore, India

ML / AI Systems Experience

testAIing

Jan–Jul 2026

Machine Learning Intern

Remote

- Owned ML R&D detecting AI-generated/deepfake speech (ECAPA-TDNN, WavLM, mixture-of-experts); achieved **0.94–0.96 AUC** on OOD benchmarks and **1.97% EER** after Hindi fine-tuning. [blog]
- Productionized inference with FastAPI, Docker and Kubernetes: HPA/KEDA scaling, probes, rolling deploy/rollback, Prometheus/Grafana; implemented Playwright/Xvfb meeting capture + WhisperX/Pyannote pipeline. **1st Prize/70+ interns (research presentation competition)**.

Samsung Research India

May–Jul 2025

Software Engineering Intern, Server Platform / MAGPIE

Bangalore

- Designed and built a multi-agent root-cause analysis system** with Google ADK/MCP and tool surfaces for Kubernetes, Prometheus, Jaeger, OpenSearch and Milvus; added scoped context, cross-incident memory, hypothesis validation and guarded halting/reporting.
- Built an OpenTelemetry + Chaos Mesh evaluation environment against a **90-scenario** microservice fault-injection benchmark; achieved strong Top-1 RCA on an internal dev slice, earning a full-time **PPO**.

Samsung R&D Institute Bangalore – PRISM

Oct 2024–May 2025

Team Lead / Co-primary Researcher

Hybrid

- Co-led 4-person **HOMA**, an SLM-coordinated multilingual smart-home agent system for decomposing multi-device commands into sequential/concurrent execution across **11+ languages**; benchmarked Gemma/Qwen/Phi-family models for edge constraints. **Samsung PRISM Excellent Worklet Award**; presented at ICCNT 2025. [code]

CEEW

Jun–Aug 2024

Data Science Intern

Delhi

- Built a BERTopic (transformer/UMAP/HDBSCAN) pipeline over **93K skill segments**; prototype informed the public **CEEW–ADB SMART** platform and a 2026 CEEW workforce report. [HF] [SMART] [report]
- Co-developed Delhi air-quality bias correction (PM2.5 MAPE **43%→27%**, RMSE **73→42** $\mu\text{g}/\text{m}^3$), coauthored as an ICAQM 2024 poster; built a modular Python/Streamlit solar+BESS dispatch + techno-economic tool. [solar]

Education

Vellore Institute of Technology

2022–2026

B.Tech. Computer Science & Engineering (IoT)

Vellore, India

CGPA **9.23/10**; **Top 1%** nationally, NPTEL Reinforcement Learning (IITM) – invited to interview for RL Instructor, IITM BS; **Top 5%** Responsible & Safe AI (IIITH × IITM). **Final Year Project:** SOPRA (see below).

Research & Applied ML

CA-AMFA | Online adaptation, IoT security, Raspberry Pi

[paper] [code]

- First-author with Dr. Yokes Babu S; IEEE **Best Paper**; adaptive authentication prototype with Thompson Sampling reducing simulated FAR **44.75%** vs fixed weights.

Manuscripts under review:

- BehaviorFL/BehaviorFL-MT**, privacy-preserving federated in-vehicle recommendation under heterogeneous sensors & partial supervision, co-inventor (Published Patent App. **202641089937**); discovery-aware sampling gives **5.8×** rare-task exposure at fixed budget. *NeurIPS 2026 NewInML Workshop*.
- SOPRA**, LLM Java test repair, supervised by Dr. Sathya K; **58.7% EM (+14.7 pp)**. *SANER 2027*.
- HERO** treats each full charging-station layout as a bandit arm for SUMO-in-the-loop placement search; **+39.9%** composite reward vs. best baseline across 13 methods. *Elsevier EAAI*.
- Ranking Is Not Valuation** audits carbon-claim overstatement from post-selection bias; reporting bound cuts overstatement **51%→11%** without changing the selected product. *NeurIPS 2026 climate workshop; patent in prep*.

ML Systems & Open-Source Projects

ChatterBot | FastAPI, Ray Serve/vLLM, Redis, RabbitMQ, PostgreSQL, SSE

[code]

- End-to-end LLM-serving prototype with sync/queued/streaming modes, context caching, AWQ path, Nginx/Docker, rate limits, retries/circuit breakers and Locust load tests.

Narad + Sahayata | Agents/RAG, recommendation, voice, ADK/MCP

[Narad] [Sahayata org]

- National-winning Narad: owned personalized recommender + model adaptation in 22-language AI4Bharat/FAISS system. Sahayata: FastAPI + Gemini Live + Vertex RAG + WebSockets + Mem0-backed persistent memory for voice-first study/wellness prototype.

Maieutic + PromptTrend + BANDITS-IDS | Research tooling; online learning

[Maieutic] [PromptTrend] [PyPI] [IDS]

- Maieutic**: Socratic Claude Code skill gating autonomy via predict-then-observe logs. **PromptTrend**: Contextual prompt recommendation API. **BANDITS-IDS**: Bandit-based intrusion detection.

Earlier Research & Recognition

NTNU × BITS Pilani: Handwritten OCR Research, with Dr. Sukalpa Chanda: end-to-end handwriting detection+recognition on IAM (ResNet-50, FOTS, rotated ROI, PHOSC recognizer); **6–8% CER** on internal runs.

VIT × King Saud University: designed stochastic ML pipeline for EV SoC prediction from CAN/BMS telemetry, advised by Dr. B. Ashok; test **MSE $\approx$ 0.041**; *Journal of Energy Storage* (IF $\approx$ 11, Q1) 97 (2024). [paper] [code]

Leadership: *Co-Founder, Team Prometheus* (30+); led zero-to-one RoboCup AI R&D, RL prototyping; HINT'25 NMPC paper [code]. *Vice Chair→Chair, Youth Red Cross VIT* (750+ donations).

Hackathons: 2x National Winner (IISF Space Hackathon [code]; Samsung PRISM [code]); Top-3 KSP Datathon [code].

ML/Research: PyTorch, TensorFlow, HF, vLLM, LLMs/agents, RAG, RL/RLHF, FL. **Systems**: Docker/K8s, FastAPI, MCP/ADK, Redis/PostgreSQL/Milvus, Prometheus/Grafana. **Languages**: Python, JS/TS, Java, C++.